

Load me at

1LXI H,2050

2MOV A,M

3

4ADD A

5

6MOV B, A

7ADD A

8ADD A

9ADD B

10INX H

11ADD M

12INX H

13MOV M,A

14HLT

DataStackKeyPad

Start2050

Address (Hex)	Address	Data
0802	2050	34
0803	2051	0
0804	2052	84
0805	2053	0
0806	2054	0
0807	2055	0
0808	2056	0
0809	2057	0
080A	2058	0
080B	2059	0
080C	2060	0
080D	2061	0
080E	2062	0
080F	2063	0

Line No

Assembler Message

0Program assembled successf

Load me at

1LXI H,8050

2MOV B,M

3MVI D,01H

4FACT: CALL MUL

5DCR B

6JNZ FACT

7INX H

8MOV M,D

9HLT

10MUL: MOV E,B

11XRA A

12ML: ADD D

13DCR E

14JNZ ML

15MOV D,A

16RET

17HLT

DataStackKeyPadMen

Start8050

Address (Hex)	Address	Data
1F72	8050	5
1F73	8051	120
1F74	8052	0
1F75	8053	0
1F76	8054	0
1F77	8055	0
1F78	8056	0
1F79	8057	0
1F7A	8058	0
1F7B	8059	0
1F7C	8060	0
1F7D	8061	0
1F7E	8062	0
1F7F	8063	0

Line No

Assembler Message

0Program assembled successfully

Load me at

1LXI H,8050

2MOV B,M

3MVI D,01H

4FACT: CALL MUL

5DCR B

6JNZ FACT

7INX H

8MOV M,D

9HLT

10MUL: MOV E,B

11XRA A

12ML: ADD D

13DCR E

14JNZ ML

15MOV D,A

16RET

17HLT

DataStackKeyPadMen

Start8050

Address (Hex)	Address	Data
1F72	8050	5
1F73	8051	120
1F74	8052	0
1F75	8053	0
1F76	8054	0
1F77	8055	0
1F78	8056	0
1F79	8057	0
1F7A	8058	0
1F7B	8059	0
1F7C	8060	0
1F7D	8061	0
1F7E	8062	0
1F7F	8063	0

Line No

Assembler Message

0Program assembled successfully

Load me at

1 LXI H, 6000

2 MOVA, M

3 INX H

4 MOV B, M

5 LOOP: CMP B

6 JZ STORE

7 JC EXG

8 SUB B

9 JMP LOOP

10 EXG: MOV C, B

11 MOV B, A

12 MOV A, C

13 JMP LOOP

14 STORE: STA 6009

15 HLT

DataStackKeyPad

Start6000

Address (Hex)	Address	Data
1770	6000	38
1771	6001	19
1772	6002	0
1773	6003	0
1774	6004	0
1775	6005	0
1776	6006	0
1777	6007	0
1778	6008	0
1779	6009	19
177A	6010	0
177B	6011	0
177C	6012	0

Line No

Assembler Message

2Invalid Opcode or Pseudo op

0Assembling Aborted

Load me at

1 LXI H, 8000

2 MOV C, M

3 MVI B, 00

4 INX H

5 MOV B, M

6 CMA

7 MOV E, A

8 MVI D, 00FH

9 MOV A, B

10 CMA

11 MOV D, A

12 INX D

13 LXI H, 0000

14 NEXT: DAD B

15 SHLD 8010

16 LOOP: DAD D

17 JNC SKIP

18 MOV A, H

19 ORA L

20 JZ EXIT

21 JMP LOOP

22 SKIP: LHLD 8010

23 JMP NEXT

24 EXIT: LHLD 8010

25 HLT

DataStackKeyPad

Start8000

Address (Hex)	Address	Data
1F40	8000	60
1F41	8001	45
1F42	8002	0
1F43	8003	0
1F44	8004	0
1F45	8005	0
1F46	8006	0
1F47	8007	0
1F48	8008	0
1F49	8009	0
1F4A	8010	4
1F4B	8011	90
1F4C	8012	0
1F4D	8013	0

Line No

Assembler Message

0Program assembled successfully

Load me at

1 LXIH, 8050

2 MOV C, M

3 INXH

4 MOV B, M

5 DCRC

6 LOOP: INXH

7 MOVA, M

8 CMPB

9 JNC SKIP

10 MOV B, A

11 SKIP: DCRC

12 JNZ LOOP

13 LXIH, 8500

14 MOV M, B

15 HLT

DataStackKeyPad

Start8050

Address (Hex)	Address	Data
1F72	8050	5
1F73	8051	2
1F74	8052	4
1F75	8053	5
1F76	8054	7
1F77	8055	9
1F78	8056	0
1F79	8057	0
1F7A	8058	0
1F7B	8059	0
1F7C	8060	0
1F7D	8061	0
1F7E	8062	0

Line No

Assembler Message

1Invalid Opcode or Pseudo op

0Assembling Aborted

Load me at

1 LDA 2001

2 MOV B,A

3 MVI C,#01

4 MVI E,#01

5 LOOP: MOV D,C

6 MVI A,00H

7 LP: ADD E

8 DCR D

9 JNZ LP

10 MOV E,A

11 INR C

12 DCR B

13 JNZ LOOP

14 MOV A,E

15 STA 2010

16 HLT

DataStackKeyPadMemoryI/O Ports

Start2001OK

Address (Hex)	Address	Data
07D1	2001	2
07D2	2002	6
07D3	2003	3
07D4	2004	2
07D5	2005	5
07D6	2006	0
07D7	2007	0
07D8	2008	0
07D9	2009	0
07DA	2010	2
07DB	2011	0
07DC	2012	0
07DD	2013	0
07DE	2014	0

Line No

Assembler Message

0Program assembled successfully

Load me at

1 LXI H,8000

2 MOV C,M

3 DCR C

4 LOOP3: MOV D,C

5 LXI H,8001

6 LOOP2: MOV A,M

7 INX H

8 CMP M

9 JC LOOP1

10 MOV B,M

11 MOV M,A

12 DCX H

13 MOV M,B

14 INX H

15 LOOP1: DCR D

16 JNZ LOOP2

17 DCR C

18 JNZ LOOP3

19 HLT

DataStackKeyPadMemoryI/O Ports

Start8000

Address (Hex)	Address	Data
1F40	8000	3
1F41	8001	0
1F42	8002	4
1F43	8003	18
1F44	8004	0
1F45	8005	0
1F46	8006	0
1F47	8007	0
1F48	8008	0
1F49	8009	0
1F4A	8010	0
1F4B	8011	0
1F4C	8012	0
1F4D	8013	0

Line No

Assembler Message

0Program assembled successfully

Load me at

1 MVI A,9AH

2 ANI 80H

3 JZ NEG

4 MVI A,22

5 JMP ST0

6 NEG: MVI A,11

7 ST0: STA 8501

8 HLT

Start8050

Address (Hex)	Address	Data
1F72	8050	0
1F73	8051	22
1F74	8052	0
1F75	8053	0
1F76	8054	0
1F77	8055	0
1F78	8056	0
1F79	8057	0
1F7A	8058	0
1F7B	8059	0
1F7C	8060	0
1F7D	8061	0
1F7E	8062	0
1F7F	8063	0

Line No

Assembler Message

0Program assembled successfully

Load me at

```
1 LDA 8050H
2 ANI 01
3 JZ LOOP1
4 MVI A,11
5 JMP LOOP2
6 LOOP1: MVI A,22
7 LOOP2: STA 8051
8 HLT
```

Data Stack KeyPad **Memory** I/O Port

Start 8050

Address (Hex)	Address	Data
---------------	---------	------

1F72	8050	20
1F73	8051	22
1F74	8052	0
1F75	8053	0
1F76	8054	0
1F77	8055	0
1F78	8056	0
1F79	8057	0
1F7A	8058	0
1F7B	8059	0
1F7C	8060	0
1F7D	8061	0
1F7E	8062	0
1F7F	8063	0

Line No Assembler Message

0 Program assembled successfully

Load me at

```
1 LDA 8000
2 CMA
3 STA 8001
4 HLT
```

Start 8000

Address (Hex)	Address	Data
---------------	---------	------

1F40	8000	6
1F41	8001	249
1F42	8002	0
1F43	8003	0
1F44	8004	0
1F45	8005	0
1F46	8006	0
1F47	8007	0
1F48	8008	0
1F49	8009	0
1F4A	8010	3
1F4B	8011	0
1F4C	8012	0
1F4D	8013	0

Line No Assembler Message

0 Program assembled successfully

Load me at

```
1 LDA 1100
2 MOV B, A
3 LDA 1101
4 MOV C, A
5 STA 1102
6 MOV A, B
7 STA 1103
8 HLT
9
```

Data Stack KeyPad **Memory**

Start 1100

Address (Hex)	Address	Data
---------------	---------	------

044C	1100	6
044D	1101	4
044E	1102	4
044F	1103	6
0450	1104	0
0451	1105	0
0452	1106	0
0453	1107	0
0454	1108	0
0455	1109	0
0456	1110	0
0457	1111	0
0458	1112	0
0459	1113	0

Line No Assembler Message

0 Program assembled successfully

Load me at

```
1 LXI H,8000H
2 MOV B,M
3 INX H
4 MOV A,M
5 CMP B
6 JC STORE
7 MOV A,B
8 STORE: STA 8010H
9 HLT
```

Start 8000

Address (Hex)	Address	Data
1F40	8000	10
1F41	8001	0
1F42	8002	0
1F43	8003	0
1F44	8004	0
1F45	8005	0
1F46	8006	0
1F47	8007	0
1F48	8008	0
1F49	8009	0
1F4A	8010	3
1F4B	8011	0
1F4C	8012	0
1F4D	8013	0

Line No Assembler Message

0 Program assembled successfull

Load me at

```
1 LDA 2050
2 MOV B,A
3 LDA 2051
4 CMP B
5 JNC STORE
6 MOV A,B
7 STORE: STA 2052
8 HLT
```

Data Stack KeyPad Memory

Start 2050

Address (Hex)	Address	Data
0802	2050	29
0803	2051	22
0804	2052	29
0805	2053	0
0806	2054	0
0807	2055	0
0808	2056	0
0809	2057	0
080A	2058	0
080B	2059	0
080C	2060	0
080D	2061	0
080E	2062	0
080F	2063	0

Line No Assembler Message

0 Program assembled successfully

8085 Microprocessor Simulator

File | Reset | Assembler | Debug | Help

Registers: A 00, BC 00 00, DE 00 02, HL 00 03, PSW 00 00, PC 42 22, SP 00 0A, Int-Reg 00

Flag: S 0, Z 1, AC 0, P 1, C 0

Load me at: []

1 LHL 2050
2 SPHL
3 LHL 2052
4 XCHG
5 LXI H, 0000H
6 LXI B, 0000H
7 AGAIN: DAD SP
8 JNC START
9 INX B
10 START: DCX D
11 MOV A, E
12 ORA D
13 JNZ AGAIN
14 SHLD 2054
15 MOV L, C
16 MOV H, B
17 SHLD 2055
18 HLT

Decimal - Hex Conversion: Decimal 0, Hex 0, To Hex, To Dec

I/O Ports: 0, Update Port Value

Memory: 0, Update Memory

Data Stack KeyPad Memory

Start 2050

Address (Hex)	Address	Data
0802	2050	10
0803	2051	0
0804	2052	5
0805	2053	0
0806	2054	50
0807	2055	0
0808	2056	0
0809	2057	0
080A	2058	0
080B	2059	0
080C	2060	0
080D	2061	0
080E	2062	0
080F	2063	0

Line No Assembler Message
0 Program assembled successfully

GNUSim8085 - 8085 Microprocessor Simulator

File | Reset | Assembler | Debug | Help

Registers: A 00, BC 00 00, DE 00 02, HL 00 03, PSW 00 00, PC 42 14, SP FF FF, Int-Reg 00

Flag: S 0, Z 1, AC 0, P 1, C 0

Load me at: []

1 LHL 2050
2 XCHG
3 LHL 2052
4 MVI C, 00
5 MOV A, E
6 SUB L
7 STA 2054
8 MOV A, D
9 SUB H
10 STA 2055
11 HLT

Decimal - Hex Conversion: Decimal 0, Hex 0, To Hex, To Dec

I/O Ports: 0, Update Port Value

Memory: 0, Update Memory

Data Stack KeyPad Memory I/O Ports

Start 2050

Address (Hex)	Address	Data
0802	2050	2
0803	2051	0
0804	2052	3
0805	2053	0
0806	2054	255
0807	2055	0
0808	2056	0
0809	2057	0
080A	2058	0
080B	2059	0
080C	2060	0
080D	2061	0
080E	2062	0
080F	2063	0

Line No Assembler Message
0 Program assembled successfully

Simulator: Idle

File | Reset | Assembler | Debug | Help

Registers: A 0A, BC 05 03, DE 00 00, HL 00 00, PSW 00 00, PC 42 17, SP FF FF, Int-Reg 00

Flag: S 0, Z 0, AC 0, P 1, C 0

Load me at: []

1 LDA 3050
2 MOV B, A
3 LDA 3051
4 ADD B
5 STA 3052
6 LDA 3053
7 MOV B, A
8 LDA 3054
9 ADC B
10 STA 3055
11 HLT

Decimal - Hex Conversion: Decimal 0, Hex 0, To Hex, To Dec

I/O Ports: 0, Update Port Value

Memory: 0, Update Memory

Data Stack KeyPad Memory

Start 3050

Address (Hex)	Address	Data
0BEA	3050	2
0BEB	3051	3
0BEC	3052	5
0BED	3053	5
0BEE	3054	5
0BEF	3055	10
0BF0	3056	0
0BF1	3057	0
0BF2	3058	0
0BF3	3059	0
0BF4	3060	0
0BF5	3061	0
0BF6	3062	0
0BF7	3063	0

Line No Assembler Message
0 Program assembled successfully

Registers	Flag
A 03	S 1
BC 02 03	Z 0
DE 00 00	AC 0
HL 00 00	P 0
PSW 00 00	C 1
PC 42 1B	
SP FF FF	
PC-Reg 00	

Decimal - Hex Conversion

Decimal	Hex
0	0

I/O Ports

00

Memory

00

Load me at:

```
1 START: NOP
2
3 LDA 8500
4 MOV B,A
5
6 LDA 8501
7 MVI C,00H
8
9 LOOP: CMP B
10 JC END
11
12 SUB B
13 INR C
14 JMP LOOP
15
16
17 END: STA 8502
18 MOV A,C
19 STA 8503
20
21 HLT
```

Data		
Start	8500	
Address (Hex)	Address	Data
2134	8500	2
2135	8501	6
2136	8502	0
2137	8503	3
2138	8504	0
2139	8505	0
213A	8506	0
213B	8507	0
213C	8508	0
213D	8509	0
213E	8510	0
213F	8511	0
2140	8512	0
2141	8513	0

Line No	Assembler Message
0	Program assembled successfully

Registers	Flag
A 02	S 0
BC 00 00	Z 1
DE 00 04	AC 0
HL 00 08	P 1
PSW 00 00	C 0
PC 42 16	
SP FF FF	
PC-Reg 00	

Decimal - Hex Conversion

Decimal	Hex
0	0

I/O Ports

00

Memory

00

Load me at:

```
1 LDA 2200
2 MOV E,A
3 MVI D,00
4 LDA 2201
5 MOV C,A
6 LXI H,0000
7 BACK: DAD D
8 DCR C
9 JNZ BACK
10 SHLD 2202
11 HLT
```

Data		
Start	2200	
Address (Hex)	Address	Data
0898	2200	4
0899	2201	2
089A	2202	8
089B	2203	0
089C	2204	0
089D	2205	0
089E	2206	0
089F	2207	0
08A0	2208	0
08A1	2209	0
08A2	2210	0
08A3	2211	0
08A4	2212	0
08A5	2213	0

Line No	Assembler Message
0	Program assembled successfully

Registers

FF	S	1
05 00	Z	0
00 00	AC	0
42 0C	P	1
FF FF	C	1

Decimal - Hex Conversion

DecimalHex

0

To HexTo Dec

I/O Ports

- + 00

Update Port Value

Memory

- + 00

Update Memory

Load me at

1 LDA 8000
2 MOV B,A
3 LDA 8001
4 SUB B
5 STA 8002
6 RST 1

DataStackKeyPadMemory

Start8000

Address (Hex)	Address	Data
1F40	8000	5
1F41	8001	4
1F42	8002	255
1F43	8003	0
1F44	8004	0
1F45	8005	0
1F46	8006	0
1F47	8007	0
1F48	8008	0
1F49	8009	0
1F4A	8010	0
1F4B	8011	0
1F4C	8012	0
1F4D	8013	0

Line NoAssembler Message
0Program assembled successfully

Registers

A 07	S 0
BC 03 00	Z 0
DE 00 00	AC 0
HL 00 00	P 0
PSW 00 00	C 0
PC 42 0C	
SP FF FF	
Int-Reg 00	

Decimal - Hex Conversion

DecimalHex

00

To HexTo Dec

I/O Ports

0 - + 00

Update Port Value

Memory

0 - + 00

Update Memory

Load me at

1 LDA 8050
2 MOV B,A
3 LDA 8051
4 ADD B
5 STA 8052
6 HLT

DataStackKeyPadMemory

Start8050

Address (Hex)	Address	Data
1F72	8050	3
1F73	8051	4
1F74	8052	7
1F75	8053	0
1F76	8054	0
1F77	8055	0
1F78	8056	0
1F79	8057	0
1F7A	8058	0
1F7B	8059	0
1F7C	8060	0
1F7D	8061	0
1F7E	8062	0
1F7F	8063	0

Line NoAssembler Message
0Program assembled successfully